Note and Formula of DDA4210

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May 6, 2026

1 Lec1: introduction

There are no formulas in Lecture 1. Good luck with the rest!

2 Lec2: Advanced Ensemble Learning

2.1 Gradient Boosting:

Boosting is an ensemble technique that combines multiple weak learners to create a strong learner. The main idea is to train models sequentially, with each model focusing on the errors made by the previous ones. (用人话说，就是一波接一波地训练模型，每一波都专注于纠正前一波的错误，从而逐步提升整体的预测能力。)

- **Final Model:**

$$H_T(\mathbf{x}) = \sum_{t=1}^{T} \alpha_t h_t(\mathbf{x})$$

  - The weighted sum of all weak learners(加权和).
  - $H_T(\mathbf{x})$ is the final model after T rounds.
  - $h_t(\mathbf{x})$ is the t-th weak learner.
  - α_t is the weight of the t-th weak learner.

- **Loss Function:**

$$\mathcal{L}(H) := \frac{1}{n} \sum_{i=1}^{n} \ell(H(\mathbf{x}_i), y_i)$$

  - **Each Step:** We want to add a function h to minimize the loss as fast as possible. Using first-order Taylor expansion(一阶泰勒展开):

$$\mathcal{L}(H + \alpha h) \approx \mathcal{L}(H) + \alpha \langle \nabla \mathcal{L}(H), h \rangle$$

  - **Find h:** Minimize $\langle \nabla \mathcal{L}(H), h \rangle$, i.e.:

$$h = \arg\min_{h \in \mathbb{H}} \sum_{i=1}^{n} \frac{\partial \mathcal{L}}{\partial [H(\mathbf{x}_i)]} h(\mathbf{x}_i)$$

    * Here $\frac{\partial \mathcal{L}}{\partial [H(\mathbf{x}_i)]}$ is the gradient of the loss function w.r.t. the current model's output on the i-th sample.

∗ We train h to fit these **negative gradients(负梯度)**.

- **GBR with squared error loss:** For regression tasks with squared error loss:

$$\ell(H(\mathbf{x}_i), y_i) = \sum_{i=1}^{n} (y_i - H(\mathbf{x}_i))^2$$

  - Solve $h_{t+1} = \arg\min_{h \in \mathbb{H}} \sum_{i=1}^{n} q_i h(\mathbf{x}_i)$, where $q_i = \frac{\partial \mathcal{L}}{\partial [H(\mathbf{x}_i)]}$
  - Let $\sum_{i=1}^{n} h^2(\mathbf{x}_i) = \text{constant}$ (we can always normalize the predictions(对结果归一化处理)) and replace q_i with $-2r_i$. We have

$$\begin{aligned} h_{t+1} &= \arg\min_{h \in \mathbb{H}} \sum_{i=1}^{n} q_i h(\mathbf{x}_i) \\ &= \arg\min_{h \in \mathbb{H}} -2 \sum_{i=1}^{n} r_i h(\mathbf{x}_i) \\ &= \arg\min_{h \in \mathbb{H}} \sum_{i=1}^{n} \left(r_i^2 - 2 r_i h(\mathbf{x}_i) + (h(\mathbf{x}_i))^2 \right) \\ &= \arg\min_{h \in \mathbb{H}} \sum_{i=1}^{n} (h(\mathbf{x}_i) - r_i)^2 \end{aligned} \tag{1}$$

  - We train h_{t+1} to predict r_i, which are from the old model H_t.
  - The gradient is:

$$\frac{\partial \mathcal{L}}{\partial [H(\mathbf{x}_i)]} = -2(y_i - H(\mathbf{x}_i))$$

  - So we fit h to the residuals:

$$r_i = y_i - H(\mathbf{x}_i)$$

  - Update the model:

$$H_t(\mathbf{x}) = H_{t-1}(\mathbf{x}) + \alpha_t h_t(\mathbf{x})$$

- **GBR with Absolute loss (更鲁棒):** For regression tasks with absolute loss:
  - Square loss is easy to deal with mathematically but not robust to outliers.
  - Absolute loss (more robust to outliers):

$$\ell(y, \hat{y}) = |y - \hat{y}|$$

  - The gradient is $q_i = \frac{\partial \mathcal{L}}{\partial H(\mathbf{x}_i)} = -\text{sign}(y_i - H(\mathbf{x}_i))$.
  - Then fit h on $-q_i$, $i = 1, 2, \ldots, n$. (no longer the residuals, different from using the squared loss)

- **GBR with Huber loss(更更鲁棒):**
  - Huber loss (more robust to outliers):

$$\ell(y, \hat{y}) = \begin{cases} \frac{1}{2}(y - \hat{y})^2 & |y - \hat{y}| \le \delta \\ \delta(|y - \hat{y}| - \delta/2) & |y - \hat{y}| > \delta \end{cases}$$

  - The gradient is:

$$\frac{\partial \mathcal{L}}{\partial H(\mathbf{x}_i)} = \begin{cases} -(y_i - H(\mathbf{x}_i)) & |y_i - H(\mathbf{x}_i)| \le \delta \\ -\delta \operatorname{sign}(y_i - H(\mathbf{x}_i)) & |y_i - H(\mathbf{x}_i)| > \delta \end{cases}$$

- **GBM for classification:**
  - Predict probability of K classes:

$$p_k(\mathbf{x}) = \frac{\exp(h^{(k)}(\mathbf{x}))}{\sum_{c=1}^{K} \exp(h^{(c)}(\mathbf{x}))} \triangleq \hat{y}^{(k)}, \quad k = 1, 2, \ldots, K$$

  - Loss: $\mathcal{L}(H) = \sum_{i=1}^{n} \ell(\mathbf{y}_i, \hat{\mathbf{y}}_i)$ (e.g. cross-entropy or KL divergence)
  - Initialize $H^{(1)}, H^{(2)}, \ldots, H^{(K)}$, iterate until converge or reach max T:
    1. Calculate negative gradients for every class:

$$-g_k(\mathbf{x}_i) = -\frac{\partial \mathcal{L}}{\partial [H^{(k)}(\mathbf{x}_i)]}, \quad i = 1, \ldots, n, \ k = 1, \ldots, K$$

    2. Fit $h^{(k)}$ to $-g_k(\mathbf{x}_i)$ (负梯度), $k = 1, 2, \ldots, K$.
    3. Update: $H^{(k)} \leftarrow H^{(k)} + \alpha h^{(k)}$, $k = 1, 2, \ldots, K$.

2.2 AdaBoost

A special case of gradient boosting with exponential loss.

- Exponential loss (learns α adaptively):

$$\mathcal{L}(H) = \sum_{i=1}^{n} e^{-y_i H(\mathbf{x}_i)}$$

- Gradient: $q_i = \frac{\partial \mathcal{L}}{\partial H(\mathbf{x}_i)} = -y_i e^{-y_i H(\mathbf{x}_i)}$
- Let $w_i = \frac{1}{Z} e^{-y_i H(\mathbf{x}_i)}$, where $Z = \sum_{i=1}^{n} e^{-y_i H(\mathbf{x}_i)}$ (constant w.r.t h), so $\sum_{i=1}^{n} w_i = 1$.
  w_i is the relative contribution of $(\mathbf{x}_i, y_i)$ to the overall loss.
- Binary classification: $y \in \{-1, +1\}$, $h(\mathbf{x}) \in \{-1, +1\}$.
- Derivation:

$$\begin{aligned} h &= \arg\min_{h \in \mathbb{H}} \sum_{i=1}^{n} q_i h(\mathbf{x}_i) \\ &= \arg\min_{h \in \mathbb{H}} -\sum_{i=1}^{n} w_i y_i h(\mathbf{x}_i) \\ &= \arg\min_{h \in \mathbb{H}} \sum_{i: h(\mathbf{x}_i) \neq y_i} w_i - \sum_{i: h(\mathbf{x}_i) = y_i} w_i \\ &= \arg\min_{h \in \mathbb{H}} \sum_{i: h(\mathbf{x}_i) \neq y_i} w_i \end{aligned}$$

  * Last equality holds because $\sum_{i=1}^{n} w_i = 1$.

- Result:

$$h = \arg\min_{h \in \mathbb{H}} \sum_{i:h(\mathbf{x}_i) \neq y_i} w_i$$

$$\epsilon := \sum_{i:h(\mathbf{x}_i) \neq y_i} w_i$$

  is the weighted classification error(加权错误率).

  Note: misclassified points by H get larger weights(分类错误的点会得到更大的权重).

- Given h, find α via:

$$\alpha = \arg\min_{\alpha} \mathcal{L}(H + \alpha h) = \arg\min_{\alpha} \sum_{i=1}^{n} e^{-y_i(H(\mathbf{x}_i) + \alpha h(\mathbf{x}_i))}$$

- Differentiate w.r.t α and set to zero:

$$\sum_{i=1}^{n} y_i h(\mathbf{x}_i) e^{-(y_i H(\mathbf{x}_i) + \alpha y_i h(\mathbf{x}_i))} = 0$$

  It follows that:

$$\sum_{i:h(\mathbf{x}_i)y_i=1} e^{-(y_i H(\mathbf{x}_i) + \alpha y_i h(\mathbf{x}_i))} - \sum_{i:h(\mathbf{x}_i)y_i=-1} e^{-(y_i H(\mathbf{x}_i) + \alpha y_i h(\mathbf{x}_i))} = 0$$

$$\sum_{i:h(\mathbf{x}_i)y_i=1} w_i e^{-\alpha} - \sum_{i:h(\mathbf{x}_i)y_i=-1} w_i e^{\alpha} = 0$$

  We have $(1-\epsilon)e^{-\alpha} - \epsilon e^{\alpha} = 0$, $e^{2\alpha} = \frac{1-\epsilon}{\epsilon}$, and get:

$$\boxed{\boldsymbol{\alpha = \frac{1}{2} \ln \frac{1-\epsilon}{\epsilon}}}$$

2.3 Mixture of Experts(MoE)

A machine learning technique where multiple expert learners (e.g. neural networks) are used to divide a problem space into homogeneous regions (distinct subtasks). (用人话 (AI) 说就是，把一个复杂的问题拆分成多个子任务，每个子任务由一个专家模型来处理，从而提升整体的学习效果。)
(骗你的人话也没看懂。。)

2.3.1 The First Attempt

- Error function:

$$E = \left\| y - \sum_{j=1}^{k} g_j O_j \right\|^2$$

- y: target vector; O_j: output of expert j; g_j: proportional contribution of expert j.
- *This error function does not ensure localisation of experts.(人话：专家没有明确的分工).

2.3.2 The Second Attempt

- Error function:
$$E = \sum_{j=1}^{k} g_j \|y - O_j\|^2$$
- The system tends to devote a single expert to each training case.
- *This may not work well in practice.(人话：实际效果可能不佳)

2.3.3 The Third Attempt [Jacobs et al. 1991]

- Error function (mixture of Gaussians):
$$E_{ME} = -\log \sum_{j=1}^{k} g_j \exp\left(-\frac{1}{2}(y - O_j)^T \Sigma^{-1} (y - O_j)\right)$$
- Assume $\Sigma = I$ (Σ: Covariance matrix, 协方差矩阵), derivative w.r.t the i-th expert:
$$\frac{\partial E_{ME}}{\partial O_i} = -\left[\frac{g_i \exp\left(-\frac{1}{2}(y - O_i)^T (y - O_i)\right)}{\sum_j g_j \exp\left(-\frac{1}{2}(y - O_j)^T (y - O_j)\right)}\right](y - O_i)$$
- Compare with derivative of second attempt:
$$\frac{\partial E}{\partial O_i} = -2 g_i (y - O_i)$$
- The former considers how well expert i performs relative to others, adapting the best-fitting expert faster.

  E.g., $g_1 = 0.8$, $g_2 = 0.2$, then $\frac{0.8 \times 0.9}{0.8 \times 0.9 + 0.2 \times 0.1} \approx 0.97 > 0.8$.
  (人话：表现好的专家会得到更快的提升)

(这 nm 都是什么玩意?)

2.4 Stacking(堆叠法)

Multiple base learners' outputs $(\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_N)$ are combined by a meta-learner (stacker) to produce the final prediction $\hat{Y}$. (用人话说就是，把多个模型的预测结果再拿去训练一个新的模型，从而提升整体的预测效果。)

- Multi-level stacking: stacking can be repeated.
- Popular stackers: linear models (fast), gradient boosting (accurate).
- Base models should be diverse, expert at different data parts.
- Trade-off: accurate but slow to predict.

(为什么一讲能有这么多神秘小知识点和神秘公式？)

3 Lec3: Advanced Applications

3.1 Recommendation Systems

3.1.1 Collaborative Filtering(协同过滤)

Core idea: Use behavioral data from many users (e.g., ratings, clicks) to predict what the current user might like. (人话：大数据推荐你喜欢的东西)

- **User-Item Interaction:**
  - Explicit Feedback(显式反馈): ratings, purchases.
  - Implicit Feedback(隐式反馈): clicks, browsing time.
- **User-Item Rating Matrix:** Rows = users, columns = items, values = ratings. Typically large and sparse.
- **Distance/Similarity Measurement(相似度度量):**
  - Euclidean distance: $\text{sim}(user_i, user_j) = \frac{1}{1+\|\mathbf{x}_i - \mathbf{x}_j\|_2}$
  - Cosine similarity: $\text{sim}(user_i, user_j) = \frac{\mathbf{x}_i \cdot \mathbf{x}_j}{\|\mathbf{x}_i\|\|\mathbf{x}_j\|}$
  - Pearson correlation: $\rho_{X,Y} = \frac{\text{cov}(X,Y)}{\sigma_X \sigma_Y}$
- **Nearest-Neighbor Collaborative Filtering(最近邻协同过滤):**

  Predict utility of item i based on similar users who rated that item.
  - $\mathcal{N}$: neighborhood set (most similar users to u who rated item i)(人话：给你推荐东西的那些“邻居”用户)
  - $w_{uv} \in [0, 1]$: similarity weight between users u and v(人话：你和邻居用户的相似度)
  - Prediction:
$$\hat{x}_{ui} = \bar{x}_u + \sum_{v \in \mathcal{N}} \left((x_{vi} - \bar{x}_v) \times \frac{w_{uv}}{\sum_{v' \in \mathcal{N}} w_{uv'}} \right)$$
  - $\bar{x}_u = \frac{1}{|I_u|} \sum_{i \in I_u} x_{ui}$ (average rating of user u, where I_u is the set of items rated by u)
  - $\bar{x}_v = \frac{1}{|I_v|} \sum_{i \in I_v} x_{vi}$ (average rating of user v, where I_v is the set of items rated by v)
- **Matrix Factorization Collaborative Filtering(矩阵分解协同过滤):**
  - Notations:
    - $R = [r_{ui}] \in \mathbb{R}^{m \times n}$: incomplete user-item rating matrix
    - Ω: set of observed entries (known ratings)
    - $P = [p_1, \ldots, p_u, \ldots, p_m] \in \mathbb{R}^{f \times m}, \quad Q = [q_1, \ldots, q_i, \ldots, q_n] \in \mathbb{R}^{f \times n}$
  - Basic SVD ($R \approx P^T Q$):
$$\min_{P,Q} \sum_{(u,i) \in \Omega} \left\{ (r_{ui} - p_u^T q_i)^2 + \lambda(\|p_u\|^2 + \|q_i\|^2) \right\}$$
  - SVD with bias ($b_{ui} = \mu + b_u + b_i$):
$$\min_{P,Q,B} \sum_{(u,i) \in \Omega} \left\{ (r_{ui} - \mu - b_u - b_i - p_u^T q_i)^2 + \lambda(\|p_u\|^2 + \|q_i\|^2 + b_u^2 + b_i^2) \right\}$$

  - μ: global mean; b_u: user bias; b_i: item bias;
  - $\lambda(\cdot)$: regularization term (prevents overfitting).
  - Optimization: GD/SGD or Alternating Least Squares(交替最小二乘法).

- **Pros & Cons of Collaborative Filtering:**
  - Pros: No domain knowledge needed; captures diverse user preferences(人话：不需要领域知识，能捕捉多样的用户偏好).
  - Cons: Cold-start problem (new users/items); data sparsity(人话：新来的用户和物品数据较少，导致推荐效果差).

3.1.2 Content-Based Methods

- **Content analysis:** item $\rightarrow$ feature vector v (e.g. TF-IDF, image features).
- **Profile learning:** user $\rightarrow$ feature vector z (e.g. age, sex, education).
- **Filtering module:** train classification/regression model to predict user's utility for an item.
- **Recommendation for user:**
  - n: # of items;
  - $z_u \in \mathbb{R}^d$:
  - user u's feature vector;
  - $h : \mathbb{R}^d \rightarrow \mathbb{R}^n$ (e.g. neural network);
  - h_i: i-th output of h.
  - ℓ: loss function (e.g. squared loss).

$$\min_h \sum_{(u,i)\in\Omega} \ell(r_{ui}, h_i(z_u))$$

- **Recommendation for item:**
  - m: # of users;
  - $v_i \in \mathbb{R}^{d'}$: item
  - i's feature vector;
  - $g : \mathbb{R}^{d'} \rightarrow \mathbb{R}^m$ (e.g. neural network);
  - g_u: u-th output of g.
  - ℓ: loss function (e.g. squared loss).

$$\min_g \sum_{(u,i)\in\Omega} \ell(r_{ui}, g_u(v_i))$$

- **Pros & Cons of Content-Based Methods:**
  - Pros: User-independent; explainable; handles new items/users well.
  - Cons: Needs domain knowledge; narrow recommendations (similar items).

3.1.3 Hybrid Methods(看起来是不重要的知识点)

Most modern systems are hybrid recommenders.

- Combine separate recommenders (CF + CB): ensemble techniques (linear weighting, stacking, etc.)
- Add content-based aspects to CF: e.g. matrix factorization with side information.

3.1.4 Evaluation Metrics for RS

3.1.4.1 Prediction Metrics(评分预测指标)

- **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{|\mathcal{T}|} \sum_{(u,i)\in\mathcal{T}} |r_{ui} - \hat{r}_{ui}|$$

- **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{|\mathcal{T}|} \sum_{(u,i)\in\mathcal{T}} (r_{ui} - \hat{r}_{ui})^2}$$

Here $\mathcal{T}$ denotes the set of user–item pairs used for evaluation (e.g., the test set).(人话：测试集中所有用户和物品的组合)

3.1.4.2 Ranking-based Metrics(基于排序的指标)

- **Precision@K:** fraction of top-K recommended items that are relevant(人话：前 K 个推荐中有多少是用户喜欢 (相关的)).

$$\text{Prec}(R)_k = \frac{|\{r \in R : \ r \le k\}|}{k}$$

- **Recall@K:** fraction of relevant items covered in top-K(人话：前 K 个推荐中，用户喜欢的相关物品占用户喜欢的物品总数).

$$\text{Recall}(R)_k = \frac{|\{r \in R : \ r \le k\}|}{|R|}$$

  - (Precision = TP/(TP+FP); Recall = TP/(TP+FN)).
  - r = rank position of a recommended item;
  - k = cut-off (top-k);
  - R = set of relevant items for the user.

- **Average Precision (AP@N):** average of precision values at ranks of relevant items.(前 N 个推荐中相关物品的精确率)

$$\text{AP@}N = \frac{1}{m} \sum_{k=1}^{N} P(k) \cdot \text{rel}(k)$$

where $P(k)$ is precision@k, m number of relevant items, and rel(k) is indicator if item at rank k is relevant.

- **Mean Average Precision (MAP):** mean of AP over Q users:

$$\text{MAP} = \frac{1}{Q} \sum_{q=1}^{Q} \text{AP}(q)$$

- **Normalized Discounted Cumulative Gain (NDCG):** evaluates ranked relevance with position discounting.

$$\text{NDCG}_p = \frac{\text{DCG}_p}{\text{IDCG}_p}, \quad \text{DCG}_p = \sum_{i=1}^{p} \frac{2^{\text{rel}_i} - 1}{\log_2(i+1)}$$

where rel_i is relevance of item at rank i, and IDCG_p is the ideal DCG (sorted by relevance). Range: $[0, 1]$. (NDCG 越接近 1 越好)

  - **Example:** 5 recommended items with relevances $[3, 2, 1, 0, 2]$ (in rank order).

$$\text{DCG}_5 = \frac{2^3-1}{\log_2(1+1)} + \frac{2^2-1}{\log_2(2+1)} + \frac{2^1-1}{\log_2(3+1)} + \frac{2^0-1}{\log_2(4+1)} + \frac{2^2-1}{\log_2(5+1)} \approx 10.5538,$$

$$\text{IDCG}_5 = \text{DCG}_5(\text{sorted rel} = [3, 2, 2, 1, 0]) \approx 10.8235,$$

$$\text{NDCG}_5 = \frac{\text{DCG}_5}{\text{IDCG}_5} \approx 0.975. \quad IDCG \qquad DCG \qquad DCG$$

(这一坨又是什么玩意？)

3.2 Learning to Rank(L2R) (排序学习)

Learning to Rank (L2R) trains models to order items by relevance, optimizing ranking-specific objectives such as pairwise or listwise losses. (你说得对，但是这里好像也没有什么公式啊 (雾))

- L2R is a supervised learning problem for ranking.
- Training data consists of:
  - A set of queries $Q = \{q_1, \ldots, q_m\}$
  - A set of documents D
  - For each query i, relevant documents $D_i = \{d_{i,1}, \ldots, d_{i,n_i}\} \subseteq D$
  - Relevance scores $\mathbf{y}_i = (y_{i,1}, \ldots, y_{i,n_i})$ for each $d_{i,j}$
- Goal: Given a new query q, output a sorted list of relevant documents.
- **Point-wise Modeling:** Predicts each (query, document) pair independently.
  Pro: Simple, can use regression/classification.
  Con: Ignores relative order between documents.
  (一句话：点对点建模简单但忽略了文档间的相对顺序。)
- **Pair-wise Modeling:** Predicts preference between document pairs for a query.
  Pro: Models relative order.
  Con: Cannot distinguish excellent-bad from fair-bad pairs.
  (一句话：成对建模能捕捉相对顺序，但无法区分优秀-差和一般-差的对比。)
- **List-wise Modeling:** Predicts for the whole ranked list of documents.
  Pro: Considers position in ranking, aligns with ranking metrics.
  Con: High training complexity.
  (一句话：列表建模考虑排名位置，但训练复杂度高。)
- **Evaluation for L2R:** Use benchmark datasets and ranking metrics (e.g., MAP, NDCG).

- **Algorithms for L2R:**
  - **Example: Ranking SVM (pairwise):**
    - **Goal:** Learn a scoring function $h(x) = w^\top x$ such that for any pair with labels $y_i > y_j$ we have $h(x_i) > h(x_j)$. (人话：让相关性更高的文档得分更高)
    - **Training pairs:** Construct pair set $\mathcal{P} = \{(i,j) : y_i > y_j\}$; $m = |\mathcal{P}|$ denotes number of pairs.
    - **Optimization (primal):**

$$\min_{w,\,\xi_{ij}\geq 0} \ \frac{1}{2}\|w\|^2 + \frac{C}{m}\sum_{(i,j)\in\mathcal{P}} \xi_{ij}$$
$$\text{s.t.} \qquad w^\top x_i \geq w^\top x_j + 1 - \xi_{ij}, \quad \forall (i,j) \in \mathcal{P}.$$

    - **Notes:** ξ_{ij} are hinge-loss slacks; C controls margin vs. training error. The objective is equivalent to minimizing the average pairwise hinge loss.
    - **Prediction:** Score each document by $h(x) = w^\top x$ and sort descending to produce a ranking.
    - **Remarks:** Works well for pairwise preferences; training can be expensive due to $O(n^2)$ pairs, so sampling or stochastic methods are often used. Kernel SVMs and regularization extend naturally.
    - (人话总结：Ranking SVM 通过学习一个线性评分函数来排序文档，优化目标是最大化正确排序对的间隔 + 最小化排序错误的惩罚。训练时需要处理大量文档对，复杂度高。)

(叽里咕噜说什么在)

4 Lec4-1: Graph Cut and Spectral Clustering(谱聚类)

4.1 Graph Paritition

A similarity graph G=(V,E,W) represents data points as vertices V, with an edge in E when the pairwise similarity is positive and weights W storing those affinities. The affinity matrix records these pairwise similarities. Graph partitioning (clustering) aims to split the graph so that edges inside a group have large weights while edges across groups have small weights. (人话：图划分就是把图分成若干部分，使得每个部分内的节点之间联系紧密 (组内权重大)，而不同部分之间的联系较弱 (组间权重小)。)

Given data points, a similarity graph can be constructed using methods such as **k-nearest neighbor or ϵ-neighborhood**. The edge weights are often defined by a **Gaussian kernel**:

$$k(x_i, x_j) = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$$

4.2 Minimum Cut

- Minimum cut: partition the graph into two sets A and B minimizing $\text{cut}(A,B) := \sum_{i\in A,\, j\in B} w_{ij}$.
- Solvable efficiently (e.g. via max-flow/min-cut algorithms, typical cost O(|V||E|)), but the minimum cut often yields unbalanced solutions (it may isolate vertices). (人话：最小割可以高效求解，但结果往往不平衡，可能会把一些节点孤立出来。)

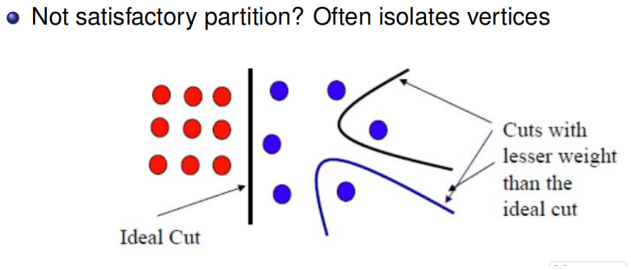


图 1: Minimum Cut Example

(画的不错的一张图)

- To address this, we can use **Normalized Cut (Ncut)**:

4.3 Normalized Cut

Normalized Cut balances cut weight with cluster sizes. For a partition (A,B), define

$$\mathrm{vol}(A) = \sum_{i \in A} d_i, \quad d_i = \sum_j w_{ij},$$

and

$$\mathrm{Ncut}(A, B) := \mathrm{cut}(A, B) \left(\frac{1}{\mathrm{vol}(A)} + \frac{1}{\mathrm{vol}(B)} \right).$$

- Minimizing Ncut favors balanced partitions but is **NP-hard**;
- **spectral clustering (谱聚类)** provides an efficient relaxation.

4.3.1 Degree Matrix and Graph Laplacian

$$W = \begin{bmatrix} w_{11} & w_{12} & \cdots & w_{1N} \\ w_{21} & w_{22} & \cdots & w_{2N} \\ \vdots & \vdots & \ddots & \vdots \\ w_{N1} & w_{N2} & \cdots & w_{NN} \end{bmatrix}$$

$$D = \begin{bmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_N \end{bmatrix}, \qquad d_j = \sum_{i=1}^{N} w_{ij}.$$

- D is the **degree matrix**. And the **graph Laplacian matrix** is defined as

$$L := D - W.$$

- Properties: L is symmetric positive semi-definite, $\mathbf{1}^\top L = 0$, and its eigenvectors are used in spectral clustering (relaxation of Ncut).

4.3.2 Normalized Cut and Graph Laplacian

Mathematical derivation (optional)

Recall $L = D - W$ and $D = \mathrm{diag}(d_1, \ldots, d_N)$.
Let $u = [u_1, u_2, \ldots, u_N]^\top$ with

$$u_i = \begin{cases} \dfrac{1}{\mathrm{vol}(A)}, & \text{if } i \in A, \\ -\dfrac{1}{\mathrm{vol}(B)}, & \text{if } i \in B. \end{cases}$$

Then

$$u^\top L u = \tfrac{1}{2} \sum_{i,j} w_{ij}(u_i - u_j)^2 = \sum_{i \in A, j \in B} w_{ij} \left(\frac{1}{\mathrm{vol}(A)} + \frac{1}{\mathrm{vol}(B)} \right)$$

and

$$u^\top D u = \sum_i d_i u_i^2 = \sum_{i \in A} \frac{d_i}{\mathrm{vol}(A)^2} + \sum_{j \in B} \frac{d_j}{\mathrm{vol}(B)^2} = \frac{1}{\mathrm{vol}(A)} + \frac{1}{\mathrm{vol}(B)}.$$

Therefore

$$\frac{u^\top L u}{u^\top D u} = \sum_{i \in A, j \in B} w_{ij} \left(\frac{1}{\mathrm{vol}(A)} + \frac{1}{\mathrm{vol}(B)} \right) = \mathrm{Ncut}(A, B).$$

(这几把都是什么玩意？好像是上面的推导吧)
(不管了，反正是 optional，摆在这图个吉利)

Conclusion

- Ncut is equivalent to minimizing the Rayleigh quotient $\dfrac{u^\top L u}{u^\top D u}$, i.e.,

$$\min_{A,B} \mathrm{Ncut}(A, B) \iff \min_u \frac{u^\top L u}{u^\top D u}, \quad u \in \mathbb{R}^N, \quad u_i = \begin{cases} \dfrac{1}{\mathrm{vol}(A)}, & i \in A, \\ -\dfrac{1}{\mathrm{vol}(B)}, & i \in B. \end{cases}$$

- Equivalent formulation: minimize the quotient subject to $u^\top D\mathbf{1} = 0$ and binary constraints $u_i \in \{1, -b\}$ (for some $b > 0$).

- Relaxation:

$$Lu = \lambda D u,$$

  taking the eigenvector corresponding to the **second smallest eigenvalue** as the relaxed solution.

- Equivalently, use the normalized Laplacian $\widetilde{L} = D^{-1}L = I - D^{-1}W$.
  Obtain a binary partition by thresholding u at 0: $i \in A$ if $u_i \geq 0$, else $i \in B$.

- Extend to k clusters by using the first k nontrivial eigenvectors and applying k-means (spectral clustering).

- (人话：2 类划分找第二小特征值对应的特征向量，直接用正负号判断。而 k 类划分用前 k 个非平凡特征向量，然后要多一步 k-means 聚类，不能直接简单的判断符号。)

(事实上还是没看懂，插个眼以后复习的时候看看有没有什么需要补的)

4.4 Spectral Clustering Algorithm

- Input: data $X = \{x_1, x_2, \ldots, x_N\}$ and number K of clusters.
- **Step 1**: Construct a similarity (affinity) matrix W (e.g. Gaussian kernel)

$$w_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$$

  and build either a k-nearest neighbor or ϵ-neighborhood graph.
- **Step 2**: Compute Laplacian L (or normalized variants):

$$L = D - W, \qquad \widetilde{L} = D^{-1}L = I - D^{-1}W, \qquad \hat{L}_{sym} = I - D^{-1/2}WD^{-1/2}.$$

- **Step 3**: Eigen-decompose (normalized) Laplacian and take first K nontrivial eigenvectors to form $Z = [v_1, \ldots, v_K]^\top \in \mathbb{R}^{K\times N}$.(对 L 做特征值分解)
- **Step 4**: Normalize the columns (row-wise embedding) to unit ℓ_2 norm(归一化):

$$z_i \leftarrow z_i/\|z_i\|, \quad i = 1, \ldots, N.$$

- **Step 5**: Run K-means on $\{z_1, \ldots, z_N\}$ and output K clusters (assignments on Z or map back to X).
- Notes: use $\widetilde{L}_{sym}$ (symmetric normalized) for best numerical stability; thresholding the second eigenvector recovers a binary partition.
- **Properties of L:**
  For $L = D - W$ or $\hat{L} = I - D^{-1/2}WD^{-1/2}$
  - L (and $\hat{L}$) are symmetric and positive semi-definite.(对称 + 半正定)
  - Eigenvalues satisfy $0 = \lambda_1 \le \lambda_2 \le \cdots \le \lambda_N$.
  - The multiplicity K of eigenvalue 0 equals the number of connected components of the graph (hence K clusters). (0 特征值的数量等于图的连通分量数，也就是聚类数)

5 Lec4-2: Semi-Supervised Learning (SSL, 半监督学习)

5.1 Notations

- Input (or feature) $\mathbf{x} \in \mathcal{X}$, output (or label) $\mathbf{y} \in \mathcal{Y}$
- Learner $f : \mathcal{X} \to \mathcal{Y}$
- Labeled data $(X_l, Y_l) = \{(\mathbf{x}_1, \mathbf{y}_1), \ldots, (\mathbf{x}_l, \mathbf{y}_l)\}$
- Unlabeled data $X_u = \{\mathbf{x}_{l+1}, \ldots, \mathbf{x}_N\}$, available during training
- Loss function $\ell : \mathcal{Y} \times \mathcal{Y} \to \mathbb{R}$
- Usually, $l \ll N$
- Test data $X_{\text{test}} = \{\mathbf{x}_{N+1}, \ldots\}$, not available during training

Why Semi-Supervised Learning?

Labeled data is often **scarce, expensive**, and requires expert annotation or specialized equipment. In contrast, unlabeled data is **abundant and cost-effective**. By leveraging both, we can build more robust and high-performing machine learning models.

5.2 Self-Training Algorithm

Assumption: One's own high confidence predictions are correct.(人话：自己高置信度的预测是正确的)

5.2.1 Algorithm Procedure:

1. Train model f from labeled data (X_l, Y_l).
2. Predict labels for unlabeled data $x \in X_u$. (用有标签数据训练模型，然后对无标签数据进行预测)
3. Add the pseudo-labeled pairs $(x, f(x))$ to the labeled training set. (把预测结果当作新的标签加入训练集)
4. Repeat the process until a stopping criterion is met.

5.2.2 Variations of Self-Training

Depending on how pseudo-labeled data is incorporated, there are several variations:

- **Most Confident:** Add only a few samples $(x, f(x))$ with the highest prediction confidence to the labeled data. (人话：只添加那些模型最有信心的预测结果)
- **All Samples:** Add all predicted $(x, f(x))$ directly to the labeled data.(直接把所有预测结果都加入训练集)
- **Weighted Approach:** Add all $(x, f(x))$ to the labeled data, but assign different weights to each sample according to its prediction confidence. (根据预测置信度给每个样本分配不同的权重)

5.2.3 Self-Training Algorithm: Propagating 1-NN (传播 1-NN)

This is a specific instance of self-training using the 1-Nearest Neighbor (1-NN) approach:

1. Classify an unlabeled sample x using the 1-NN rule (assigning the label of its nearest labeled neighbor).(用 1-NN 方法对无标签样本进行分类，分配最近的有标签邻居的标签)
2. Add the newly labeled pair $(x, f(x))$ to the labeled dataset and repeat the process.

Advantages and Disadvantages

- **Pros:** It is the simplest semi-supervised approach, serves as a versatile wrapper method, and is widely utilized in Natural Language Processing (NLP).
- **Cons:** The method is sensitive to outliers, and early classification errors can lead to self-reinforcing mistakes. (对异常值敏感)

5.3 Graph based SSL methods

Assumption: A graph is given on the labeled and unlabeled data. Instances connected by heavy edge tend to have the same label(由重边连接的实例往往具有相同的标签)
人话就是边的权重反应了两个节点 (数据点)，之间的相似程度，边权越大，数据点越相似

5.3.1 Graph Construction

- **Nodes:** The set of nodes is the union of labeled and unlabeled data, $X_l \cup X_u$.
- **Edges:** Constructed via a k-Nearest Neighbor graph (unweighted) or a fully connected graph where weights decay with distance (全连接图, 权重随距离衰减):

$$w_{ij} = \exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma^2}\right)$$

- w_{ij} 是高斯核的那个函数，反映了数据点之间的相似度

Regularized Classifier A standard classifier learns by minimizing a loss term combined with regularization (e.g., LASSO or Regularized Least Squares). We can extend this to semi-supervised learning by using unlabeled data for regularization.

Graph-Based Regularization The core principle is that if two instances x_i and x_j are similar (indicated by a large edge weight w_{ij}), their predicted labels $f(x_i)$ and $f(x_j)$ should also be similar. This leads to the following optimization problem:
其实就是学习一个学习器来使得损失函数 + 正则化项最小

$$\min_f \sum_{i=1}^{l} \ell(y_i, f(x_i)) + \lambda \sum_{i=1}^{N}\sum_{j=1}^{N} w_{ij}\|f(x_i) - f(x_j)\|^2$$

第一项：有标签数据的损失。**第二项：**图正则化项，强制相似点有相似预测。λ**：**正则化参数。

Algorithm Objectives The graph-based algorithm operates with two primary goals:

- **Label Constraint:** It enforces the correct labels on the existing labeled data.
- **Manifold Consistency:** It maximizes the consistency of unlabeled examples relative to the underlying graph topology.

5.3.2 Label Propagation Algorithm (标签传播算法)

1. Compute affinity matrix W (亲和矩阵).
2. Compute degree matrix D, where $D_{ii} = \sum_j W_{ij}$.
3. Initialize $\hat{Y}^{(0)} \leftarrow (y_1, \ldots, y_l, 0, 0, \ldots, 0)$.
4. Iterate: $\hat{Y}^{(t+1)} = D^{-1}W\hat{Y}^{(t)}$, fix labels of labeled data.
5. Repeat until convergence.
6. Assign labels based on the sign of $\hat{y}_i^{(\infty)}$.

- The algorithm forces the labels on the labeled data (强制在标记数据上保留标签)
- The algorithm tries to maximizes the consistency of the unlabeled examples with the topology of the graph (试图使未标记示例与图的拓扑结构保持最大程度的一致性)

其实到最后也不知道这一讲到底细讲了什么，并没有什么公式 (雾)

6 Lec5: GNN

6.1 Graph Data and Related Tasks

- **Graph Data**: $G = (V, E)$, $|V| = n$.
  - Vertices $V = \{v_1, \ldots, v_n\}$, Edges $E = \{e_1, \ldots, e_l\}$.
  - Affinity matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, Node feature matrix $\mathbf{X} \in \mathbb{R}^{n \times d}$.
  - Affinity matrix(亲和矩阵) , 也叫 Similarity matrix(相似矩阵)，反映了节点之间的关系
- **Tasks**:
  - **Node Embedding**: $(\mathbf{A}, \mathbf{X}) \to \mathbf{Z} \in \mathbb{R}^{n \times k}$ (Represent each node as a vector). (人话就是把每个节点表示成一个向量，具体方法这里不用深究)
  - **Graph Embedding**: $\mathcal{G} = \{G_1, \ldots, G_N\}$, $(\mathbf{A}_i, \mathbf{X}_i) \to \mathbf{g}_i \in \mathbb{R}^k$. (人话：把每个图表示成一个向量)
  - **Classification**: Node-level $(v_i \to y_i)$ or Graph-level $(G_i \to y_i)$.
  - **Others**: Link prediction, clustering, etc.
- **Traditional Embedding Methods**: Laplacian embedding, Deepwalk, LINE, node2vec.(了解就好)
- **Graph Embedding Methods**: Methods based on node embeddings, Graph kernels(了解就好)
- **Graph neural networks (GNNs)**: NNs that operate on graph-structured data.
  - **Main Idea**: Pass messages between pairs of nodes and agglomerate.
  - **Alternative Interpretation**: Pass messages between nodes to refine node (and possibly edge) representations.

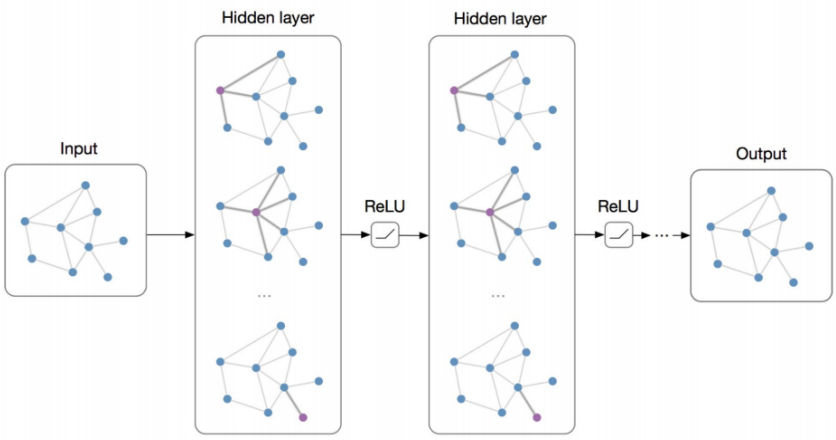


图 2: Graph Neural Networks Architecture

6.2 Graph Convolutional Networks (GCNs)

- **Graph Convolution**: $\mathbf{H}^{(l+1)} = \sigma(\widehat{\mathbf{A}}\mathbf{H}^{(l)}\mathbf{W}^{(l)})$.
  - σ: Activation function (e.g., ReLU, Sigmoid).
  - $\mathbf{W}^{(l)} \in \mathbb{R}^{d_l \times d_{l+1}}$: Parameter matrix.
  - $\widehat{\mathbf{A}} = \tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2}$, where $\tilde{\mathbf{A}} = \mathbf{A} + \mathbf{I}$, $\tilde{\mathbf{D}} = \text{diag}(\sum_j \tilde{\mathbf{A}}_{ij})$.
  - $\mathbf{H}^{(l+1)} \in \mathbb{R}^{n \times d_{l+1}}$: Output of l-th GCN layer.
  - $\mathbf{H}^{(0)} = \mathbf{X} \in \mathbb{R}^{n \times d}$: Input feature matrix.
- **Why GCNs work (optional)**: GCN is an approximated spectral convolution.
  1. $g_\theta * \mathbf{x} = g_\theta(\mathbf{L})\mathbf{x} = \mathbf{U}g_\theta(\Lambda)\mathbf{U}^\top\mathbf{x}$, where $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2} = \mathbf{U}\Lambda\mathbf{U}^\top$.
  2. Chebyshev polynomials approx: $g_\theta(\mathbf{L})\mathbf{x} \approx \sum_{k=0}^{K} \theta_k T_k(\tilde{\mathbf{L}})\mathbf{x}$, where $\tilde{\mathbf{L}} = \frac{2}{\lambda_{\max}}\mathbf{L} - \mathbf{I}$.
  3. $T_k(a) = 2aT_{k-1}(a) - T_{k-2}(a)$, with $T_0(a) = 1, T_1(a) = a$.
  4. Set $K = 1, \lambda_{\max} \approx 2 \implies g_\theta * \mathbf{x} \approx \theta_0\mathbf{x} - \theta_1\mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2}\mathbf{x}$.
  5. Assume $\theta = \theta_0 = -\theta_1 \implies g_\theta * \mathbf{x} \approx \theta(\mathbf{I} + \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2})\mathbf{x}$.
  6. Renormalization trick: $\mathbf{I} + \mathbf{D}^{-1/2}\mathbf{A}\mathbf{D}^{-1/2} \to \tilde{\mathbf{D}}^{-1/2}\tilde{\mathbf{A}}\tilde{\mathbf{D}}^{-1/2} = \hat{\mathbf{A}}$.
     (反正是 optional, 看一眼得了, 叽里咕噜也不知道在说什么)
- **Commonly used architecture**: $\mathbf{Z} = f(\mathbf{X}, \mathbf{A}) = \text{softmax}(\hat{\mathbf{A}}\text{ReLU}(\hat{\mathbf{A}}\mathbf{X}\mathbf{W}^{(0)})\mathbf{W}^{(1)})$. (注意到，只有两层)
- **Over-smoothing(过平滑)**: Deep GCNs perform poorly because graph convolution is essentially a smoothing operation.
  - Applying $\hat{\mathbf{A}}$ enough times causes node features to converge to the same value, losing discriminative information.(收敛到相同的值导致丢失判别信息)
- **The Application of GCNs**:
  - **Node Classification**: Predict labels for unlabeled nodes given partially labeled nodes.
    - Y_L: Set of labeled node indices.
    - $Y \in \{0, 1\}^{L \times K}$: Label matrix.
    - Loss (Semi-supervised): $\mathcal{L} = -\sum_{i \in Y_L} \sum_{k=1}^{K} Y_{ik} \ln Z_{ik}$.
  - **Graph Classification**: Classify a set of graphs into K categories.
    - **READOUT function**: Computes graph features from node features.(从节点特征计算图特征)
    - Sum: $\mathbf{h}_G = \sum_{i=1}^{n_G} \mathbf{h}_i$; Average: $\mathbf{h}_G = \frac{1}{n_G}\sum_{i=1}^{n_G} \mathbf{h}_i$; Min/Max pooling.
    - Loss (Supervised): $\mathcal{L} = -\sum_j \sum_{k=1}^{K} \mathbf{Y}_{jk} \ln \mathbf{Z}_{jk}$.
  - **Link Prediction**: Predict new edges in a graph G.
    - Loss: $\mathcal{L} = -\sum_{(i,j) \in \Omega}(A_{ij} \ln \sigma(\mathbf{z}_i^\top \mathbf{z}_j) + (1 - A_{ij}) \ln(1 - \sigma(\mathbf{z}_i^\top \mathbf{z}_j)))$.
    - σ is the sigmoid function.

    (这几把又是什么玩意？)

6.3 Other Graph Neural Networks(optional)

– **GraphSAGE (Inductive Learning)**: Overcomes GCN's transductive limitation by learning aggregator functions.

1. Initialize: $\mathbf{h}_v^0 \leftarrow \mathbf{x}_v, \forall v \in \mathcal{V}$.
2. For $k = 1 \dots K$:
   * Aggregate: $\mathbf{h}_{\mathcal{N}(v)}^k \leftarrow \text{AGGREGATE}_k(\{\mathbf{h}_u^{k-1}, \forall u \in \mathcal{N}(v)\})$.
   * Update: $\mathbf{h}_v^k \leftarrow \sigma\left(\mathbf{W}^k \cdot \text{CONCAT}(\mathbf{h}_v^{k-1}, \mathbf{h}_{\mathcal{N}(v)}^k)\right)$.
3. Normalize: $\mathbf{h}_v^k \leftarrow \mathbf{h}_v^k / \|\mathbf{h}_v^k\|_2$. Output: $\mathbf{z}_v \leftarrow \mathbf{h}_v^k$.

– **Graph Attention Network (GAT)**: Uses attention to assign weights to neighbors instead of fixed weights.

* Self-attention: $e_{ij} = a(\mathbf{W}\mathbf{h}_i, \mathbf{W}\mathbf{h}_j) = \text{LeakyReLU}(\mathbf{a}^\top [\mathbf{W}\mathbf{h}_i \| \mathbf{W}\mathbf{h}_j])$.
* Attention coefficient: $\alpha_{ij} = \frac{\exp(e_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(e_{ik})}$.
* Output: $\mathbf{h}_i' = \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij} \mathbf{W}\mathbf{h}_j\right)$.
* Multi-head: $\mathbf{h}_i' = \|_{m=1}^M \sigma\left(\sum_{j \in \mathcal{N}_i} \alpha_{ij}^{(m)} \mathbf{W}^{(m)} \mathbf{h}_j\right)$.

很好，依然是 optional, 依然看不懂一点

7 Lec6: Non-Linear Dimensionality Reduction

7.1 Locally Linear Embedding (LLE)

- **Core Intuition**: Based on the geometric intuition of **local linearity**. It assumes that each sample and its neighbors lie on or near a locally linear patch of a low-dimensional manifold(流形) embedded in a high-dimensional space.
- **LLE Assumption**: The projection from high to low dimensionality should preserve neighborhood relationships:
  – Projected data points should maintain the same neighbors as in the original space.(人话就是投影后的数据点应该和原空间中的数据点保持相同的邻居关系)
  – The **locally linear representation** (reconstruction weights) should be preserved.
- **Algorithm Steps**:
  1. **Find Neighbors**: For each $\mathbf{x}_i$, find its k nearest neighbors $\mathcal{N}_i^k$ ($k \geq d+1$,k 一般比 d 大一点).
  2. **Compute Weights W**: Solve for weights that reconstruct $\mathbf{x}_i$ from neighbors:

$$\mathbf{W} = \text{argmin}_{\mathbf{W}} \sum_{i=1}^N \|\mathbf{x}_i - \sum_{j \in \mathcal{N}_i^k} w_{ij} \mathbf{x}_j\|^2, \quad \text{s.t. } \forall i \sum_{j \in \mathcal{N}_i^k} w_{ij} = 1$$

  $w_{i,j}$: locally linear reconstruction weights, 局部线性重建权重，且其和为 1
  3. **Compute Projections Z**: Fix $\mathbf{W}$, find low-dimensional $\mathbf{z}_i \in \mathbb{R}^d$:

$$\mathbf{Z} = \text{argmin}_{\mathbf{Z}} \sum_{i=1}^N \|\mathbf{z}_i - \sum_{j \in \mathcal{N}_i^k} w_{ij} \mathbf{z}_j\|^2, \quad \text{s.t. } \sum \mathbf{z}_i = 0, \frac{1}{N}\mathbf{Z}\mathbf{Z}^\top = \mathbf{I}$$

- **Solution for Z**: We have:

$$\sum_{i=1}^{N} \|\mathbf{z}_i - \sum_{j \in \mathcal{N}_i^k} w_{ij}\mathbf{z}_j\|^2 = \|\mathbf{Z} - \mathbf{ZW}\|_F^2 = \text{trace}\left(\mathbf{Z}(\mathbf{I}-\mathbf{W})(\mathbf{I}-\mathbf{W}^\top)\mathbf{Z}^\top\right)$$

  The cost function is equivalent to $\text{trace}(\mathbf{Z}(\mathbf{I}-\mathbf{W})(\mathbf{I}-\mathbf{W})^\top\mathbf{Z}^\top)$. $\mathbf{Z}$ is composed of the d eigenvectors of $(\mathbf{I}-\mathbf{W})(\mathbf{I}-\mathbf{W})^\top$ corresponding to the **smallest** d **non-zero eigenvalues**.
  (z 由 $(\mathbf{I}-\mathbf{W})(\mathbf{I}-\mathbf{W}^\top)$ 的最小的 d 个非零特征值对应的特征向量组成)

- **Notes/Limitations**:
  - **Neighborhood size** k: Cannot be too small (insufficient neighbors) or too large (loses local linearity).
  - **Out-of-sample extension**: LLE cannot directly handle new data; the entire dataset must be recomputed for new samples. (人话就是 LLE 不能处理新数据, 必须重新计算整个数据集才能处理新样本)

7.2 t-distributed stochastic neighbor embedding(t-SNE)

Given high-dimensional data $\{\mathbf{x}_1, \ldots, \mathbf{x}_N\}$

- **Conditional Probability**: $p_{j|i}$ denotes the probability that $\mathbf{x}_j$ is a neighbor of $\mathbf{x}_i$:

$$p_{j|i} = \frac{\exp(-\|\mathbf{x}_i - \mathbf{x}_j\|^2/2\sigma_i^2)}{\sum_{k\neq i}\exp(-\|\mathbf{x}_i - \mathbf{x}_k\|^2/2\sigma_i^2)}, \quad p_{i|i} = 0$$

  - $p_{j|i}$: Probability of $\mathbf{x}_j$ being a neighbor of $\mathbf{x}_i$ (i,j 作为邻居的概率)
  - σ_i: the size of the neighborhood around $\mathbf{x}_i$ (邻域大小)
    (不同的数据点, $sigma_i$ 不同, 其定义了我们试图保留的邻域结构, 根据困惑度 (Perplexity) 来计算得出, t-SNE 降维最后的结果与这个值高度相关)

- **Bandwidth** σ_i: Sets the local neighborhood size, typically determined by user-defined **Perplexity**.

- **Joint Probability (Symmetry)**: To ensure $p_{ij} = p_{ji}$, define:

$$p_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad \text{s.t. } \sum_{i,j} p_{ij} = 1, p_{ii} = 0, (\quad)$$

- **Perplexity**: A measure of the effective number of neighbors:

$$\text{perp}(i) = 2^{H(p_{j|i})}, \quad H(p_{j|i}) = -\sum_{j=1}^{N} p_{j|i}\log p_{j|i}$$

  - Low perplexity $\implies$ distribution is good at predicting the sample (focusses on local structure). (人话就是低的困惑度意味着分布更好地预测样本, 关注局部结构)
  - σ_i is tuned (e.g., via bisection(二分法)) to match the target perplexity. (找到一个 σ_i ， 使得以该点为中心计算出来的 Perplexity 等于用户指定的目标 Perplexity, 二分法是一种有效的方法)
  - **Typical values: 5–50**.

- **t-SNE: Objective and Optimization**:
  - **Low-dimensional Similarity** q_{ij}: Uses Student's t-distribution (d.o.f.=1) to define similarity in embedding space $\mathbf{z}_i \in \mathbb{R}^d$:

$$q_{ij} = \frac{(1 + \|\mathbf{z}_i - \mathbf{z}_j\|^2)^{-1}}{\sum_k \sum_{l \neq k} (1 + \|\mathbf{z}_k - \mathbf{z}_l\|^2)^{-1}}$$

  - **Cost Function**: Minimize the KL-divergence between P and Q:

$$\mathcal{L} := \mathrm{KL}(P \| Q) = \sum_{i \neq j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

  - **Gradient**: Optimization via gradient descent:

$$\frac{\partial \mathcal{L}}{\partial \mathbf{z}_i} = \sum_j (p_{ij} - q_{ij})(\mathbf{z}_i - \mathbf{z}_j)(1 + \|\mathbf{z}_i - \mathbf{z}_j\|^2)^{-1}$$

  - **update**:

$$\mathbf{z}_i^{(t+1)} = \mathbf{z}_i^{(t)} - \eta \frac{\partial \mathcal{L}}{\partial \mathbf{z}_i}$$

- **Notes/Limitations**:
  - **Visualization**: Primarily used for 2D or 3D visualization.
  - **Complexity**: Computationally expensive; suitable for datasets with a few thousand points.
  - **Perplexity Sensitivity**: Results depend heavily on perplexity; different values may reveal different structures. (结果高度依赖困惑度)
  - **Out-of-sample**: Cannot handle new data directly; requires recomputation of the entire dataset.

7.3 Autoencoder, 自编码器

- **Core Idea**: A neural network that learns to copy its input to its output. It compresses input into a lower-dimensional latent space (Encoder) and then reconstructs the output (Decoder). (人话就是一个神经网络，自我学习将输入复制到输出。它将输入压缩到一个低维的潜在空间（编码器），然后再重构输出（解码器）)
- **Structure**:
  - **Encoder**: $h = f(\mathbf{x}) = \sigma(\mathbf{W}\mathbf{x} + \mathbf{b})$, where σ is an activation function (e.g., sigmoid, ReLU).
  - **Decoder**: $\hat{\mathbf{x}} = g(h) = \sigma(\mathbf{W}'h + \mathbf{b}')$.
- **Objective**: Minimize reconstruction error(最小化重建误差):

$$\min_\theta \frac{1}{2} \sum_{i=1}^{N} \|\mathbf{x}_i - g(f(\mathbf{x}_i))\|^2$$

  where θ represents all weights and biases.
- **Training**: Optimized via backpropagation and gradient descent.

- **Variants**:
  - **Stacked AE (SAE), 栈式自编码器**: Multi-layer encoder/decoder for deeper feature learning. Supports **out-of-sample extension** (unlike PCA/LLE/t-SNE). (这个东西可以直接对新数据进行编码)
  - **Convolutional AE (CAE), 卷积自编码器**: Replaces fully connected layers with convolutional layers; suitable for image data.

8 Lec7: Generative Models

8.1 Introduction

- **Generative Models:** Learn a probability distribution $p(\mathbf{x})$ from data $\mathcal{D} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ to sample/generate new instances with a similar distribution.
- **Deep Generative Models (DGMs):** Combine generative models with deep neural networks. Achieved SOTA in image/text generation (e.g., ChatGPT).
- **Types of DGMs:**
  - **AutoEncoder (AE)**
    Recap:
    An unsupervised network for identity transformation. It maps high-dimensional $\mathbf{x}$ to latent $\mathbf{z}$ via encoder f_ϕ, then reconstructs $\mathbf{x}'$ via decoder g_θ.
    Targeted loss: $\min_{\phi,\theta} \frac{1}{n} \sum_{i=1}^{n} \|\mathbf{x}_i - g_\theta(f_\phi(\mathbf{x}_i))\|^2 + \mathcal{R}(\phi, \theta)$.
  - Variational AutoEncoder (VAE)
  - Generative Adversarial Networks (GANs)
  - Normalizing Flows
  - Diffusion Models

8.2 Variational AutoEncoder (VAE)

- **Concept:** VAE transforms input $\mathbf{x}$ into a prior distribution p_z using encoder f_ϕ, then reconstructs it via decoder g_θ. (人话就是 VAE 输入了一个分布，将他映射到另一个分布上，不像 AE 直接映射到一个点上)
- **Generation:** Given trained f_ϕ and g_θ, a new sample is generated by:
  1. Sample $\mathbf{z}_i \sim p_z$.
  2. Generate $\hat{\mathbf{x}}_i = g_\theta(\mathbf{z}_i)$.
- **Objective:** Maximize the log-likelihood of real data samples $\mathbf{X} = \{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$:

$$\theta^* = \arg\max_\theta \sum_{i=1}^{n} \log p_\theta(\mathbf{x}_i)$$

  where $p_\theta(\mathbf{x}_i)$ denotes the probability of generating $\mathbf{x}_i$ given parameters θ.

- **Marginal Likelihood:** Direct computation is intractable because integrating over all $\mathbf{z}$ is impossible:

$$p_\theta(\mathbf{x}) = \int p_\theta(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int p_\theta(\mathbf{x}|\mathbf{z}) p_\theta(\mathbf{z}) d\mathbf{z}$$

  Using Bayes' theorem, $p_\theta(\mathbf{x}) = \frac{p_\theta(\mathbf{x}|\mathbf{z}) p_\theta(\mathbf{z})}{p_\theta(\mathbf{z}|\mathbf{x})}$, but the posterior $p_\theta(\mathbf{z}|\mathbf{x})$ cannot be computed.

- **Variational Inference:** Train another neural network (encoder) f_ϕ to learn $q_\phi(\mathbf{z}|\mathbf{x}) \approx p_\theta(\mathbf{z}|\mathbf{x})$.

- **Log-Likelihood Decomposition:** Introducing $q_\phi(\mathbf{z}|\mathbf{x})$, we decompose $\log p_\theta(\mathbf{x})$:

$$\begin{aligned}\log p_\theta(\mathbf{x}) &= \log \frac{p_\theta(\mathbf{x}|\mathbf{z}) p(\mathbf{z})}{p_\theta(\mathbf{z}|\mathbf{x})} = \log \frac{p_\theta(\mathbf{x}|\mathbf{z}) p(\mathbf{z}) q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z}|\mathbf{x}) q_\phi(\mathbf{z}|\mathbf{x})} \\ &= \log p_\theta(\mathbf{x}|\mathbf{z}) - \log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})} + \log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z}|\mathbf{x})}\end{aligned}$$

  Taking the expectation $\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}$:

$$\begin{aligned}\log p_\theta(\mathbf{x}) &= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x})] = \int q_\phi(\mathbf{z}|\mathbf{x}) \log p_\theta(\mathbf{x}) d\mathbf{z} \\ &= \int q_\phi(\mathbf{z}|\mathbf{x}) \log p_\theta(\mathbf{x}|\mathbf{z}) d\mathbf{z} - \int q_\phi(\mathbf{z}|\mathbf{x}) \log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p(\mathbf{z})} d\mathbf{z} + \int q_\phi(\mathbf{z}|\mathbf{x}) \log \frac{q_\phi(\mathbf{z}|\mathbf{x})}{p_\theta(\mathbf{z}|\mathbf{x})} d\mathbf{z} \\ &= \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z})) + D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z}|\mathbf{x}))\end{aligned}$$

- **Evidence Lower Bound (ELBO):** Since the KL-divergence is always non-negative ($D_{KL} \geq 0$), we have:

$$\log p_\theta(\mathbf{x}) \geq \mathcal{L}_{\phi,\theta}(\mathbf{x}) = \underbrace{\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]}_{\text{Reconstruction Loss}} - \underbrace{D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))}_{\text{KL Divergence}}$$

  where $\mathcal{L}_{\phi,\theta}(\mathbf{x})$ is the Evidence Lower Bound (or variational lower bound). It follows that:

$$\log p_\theta(\mathbf{x}) = \mathcal{L}_{\phi,\theta}(\mathbf{x}) + D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z}|\mathbf{x}))$$

- **VAE Objective:** VAE maximizes the ELBO:

$$\phi^*, \theta^* = \arg\max_{\phi,\theta} \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p(\mathbf{z}))$$

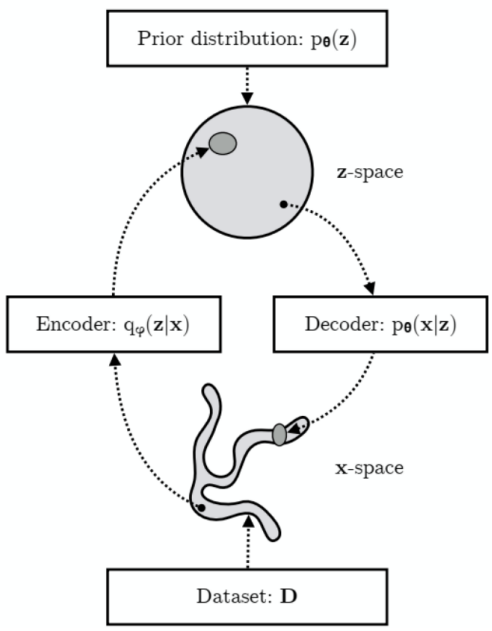
  When $\mathcal{L}_{\phi,\theta}(\mathbf{x}) = \log p_\theta(\mathbf{x})$, we have $D_{KL}(q_\phi(\mathbf{z}|\mathbf{x}) \| p_\theta(\mathbf{z}|\mathbf{x})) = 0$, meaning $q_\phi(\mathbf{z}|\mathbf{x}) = p_\theta(\mathbf{z}|\mathbf{x})$.

  (这一堆用人话说就是：由于那个 $logp_\theta(x)$ 是无法计算的，我们引入了一个新的分布 $q_\phi(z|x)$，来近似那个无法计算的后验分布 $p_\theta(z|x)$。这样得到了一个下界（ELBO），我们最大化这个下界来训练我们的模型。当这个下界等于 $logp_\theta(x)$ 的时候，我们就达到了最优，此时 $q_\phi(z|x)$ 就等于 $p_\theta(z|x)$，也就是说我们成功地近似了那个无法计算的后验分布)(并非人话，还是看不懂)
  (牛魔的这都是什么东西。)

- **Reparameterization Trick**:

  - **Problem**: During training, we need to sample $\mathbf{z}$ from $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu, \sigma^2)$ to compute the reconstruction loss. However, the sampling operation is non-differentiable (**a stochastic process**), preventing gradients from flowing back to μ and σ, which makes backpropagation impossible.

Variational Auto-Encoder (VAE): Optimization



$$\max_{\phi,\theta} \mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] - D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))$$

- maximize $\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})]$: reconstruct $\mathbf{x}$
- minimize $D_{KL}(q_\phi(\mathbf{z}|\mathbf{x})\|p(\mathbf{z}))$: approximate prior

Suppose $p(\mathbf{z}) = \mathcal{N}(\mathbf{z}; \mathbf{0}, \mathbf{I})$. Then let $q_\phi(\mathbf{z}|\mathbf{x}) = \mathcal{N}(\mu_\phi(\mathbf{x}), \text{diag}(\sigma^2_\phi(\mathbf{x})))$.

The expectation term in the loss function requires sampling $\mathbf{z} \sim q_\phi(\mathbf{z}|\mathbf{x})$, which is a stochastic process. Therefore we cannot backpropagate the gradient.

图 3: VAE Optimization and ELBO

- **Reparameterization Trick(Cont.)**:
  - **Solution:** Sample random noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$ and compute $\mathbf{z} = \mu + \sigma \odot \epsilon$.
  - **Proof of Mean:** By linearity of expectation $E[aX + b] = aE[X] + b$:

$$\begin{aligned} E[\mathbf{z}] &= E[\mu + \sigma \cdot \epsilon] = \mu + \sigma \cdot E[\epsilon] \\ &= \mu + \sigma \cdot 0 = \mu \end{aligned}$$

  - **Proof of Variance:** By variance property $Var(aX + b) = a^2 Var(X)$:

$$\begin{aligned} Var(\mathbf{z}) &= Var(\mu + \sigma \cdot \epsilon) = \sigma^2 \cdot Var(\epsilon) \\ &= \sigma^2 \cdot 1 = \sigma^2 \end{aligned}$$

  - **Result:** Randomness is shifted to ϵ. Since μ and σ are deterministic and differentiable transformations, gradients can flow from $\mathbf{z}$ to μ and σ, then back to the encoder.
- **The Details of the Loss Function**:
  - **KL Divergence Term:** Considering one element of $\mathbf{z}$:

$$\begin{aligned} D_{KL}(q_\phi(z|x)\|p(z)) &= \int \frac{1}{\sqrt{2\pi\sigma_q^2}} \exp\left(-\frac{(z-\mu_q)^2}{2\sigma_q^2}\right) \log\left(\frac{\frac{1}{\sqrt{2\pi\sigma_p^2}}\exp\left(-\frac{(z-\mu_p)^2}{2\sigma_p^2}\right)}{\frac{1}{\sqrt{2\pi\sigma_q^2}}\exp\left(-\frac{(z-\mu_q)^2}{2\sigma_q^2}\right)}\right) dz \\ &= \log\left(\frac{\sigma_q}{\sigma_p}\right) - \frac{\sigma_q^2 + (\mu_q - \mu_p)^2}{2\sigma_p^2} + \frac{1}{2} \\ &= \frac{1}{2}\left[1 + \log(\sigma_q^2) - \sigma_q^2 - \mu_q^2\right] \quad (\text{when } p(z) \sim \mathcal{N}(0,1)) \end{aligned}$$

– **Reconstruction Term:** Assuming $p_\theta(\mathbf{x}|\mathbf{z}) = \mathcal{N}(\mathbf{x}; g_\theta(\mathbf{z}), \Sigma_\mathbf{x})$:

$$\mathbb{E}_{\mathbf{z}\sim q_\phi(\mathbf{z}|\mathbf{x})}[\log p_\theta(\mathbf{x}|\mathbf{z})] \propto -\|\mathbf{x} - g_\theta(\mathbf{z})\|^2$$

(人话就是，KL 散度最后变成了那个非常简单的形式。重构损失等价于输入和输出之间的均方误差，前面那一大堆积分消失了。)
(虽然我也不知道这牛魔怎么推出来的)

8.3 Generative Adversarial Networks (GANs)

- Inspired by game theory, GAN involves an adversarial process training two networks:
  - **Generator** G: Learns the real data distribution to produce realistic samples.
  - **Discriminator** D: Estimates the probability that a sample is real rather than from G.
- **Objective**: D aims to maximize correct labeling of both real and generated samples, while G aims to maximize D's mistake probability.
  (人话就是生成器和判别器在比谁牛逼，生成器想骗过判别器，判别器想识破生成器)
- **Training Steps**:
  1. Fix G, train D to better distinguish samples.
  2. Fix D, train G to produce more realistic samples.
  3. Iterate steps 1 and 2 until convergence.
     (其实还是回合制游戏)
- **Minimax Game**:
  - **Notations**: p_r (real distribution), p_g (generator distribution), p_z (noise prior).
  - **Discriminator Objective**: Maximize $D(x)$ for $x \sim p_r$ and minimize $D(G(z))$ for $z \sim p_z$:

    $$\max_D \mathbb{E}_{\mathbf{x}\sim p_r(\mathbf{x})}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))]$$

    (判定真实数据准确并且判断生成器生成的数据不准确，提高这个概率)
  - **Generator Objective**: Minimize the probability of D correctly identifying fake samples:

    $$\min_G \mathbb{E}_{\mathbf{z}\sim p_z(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))]$$

    (生成器想让判别器判断生成的数据为真，提高这个概率)
  - **Combined Loss Function** $\mathcal{L}(G, D)$:

    $$\min_G \max_D \mathcal{V}(D, G) = \mathbb{E}_{\mathbf{x}\sim p_r(\mathbf{x})}[\log D(\mathbf{x})] + \mathbb{E}_{\mathbf{z}\sim p_z(\mathbf{z})}[\log(1 - D(G(\mathbf{z})))]$$

    (来一场痛快的回合制游戏口牙)
- **Pros & Cons**:
  - **Advantages**: Direct sampling; avoids MLE; yields more realistic samples than VAE.
  - **Disadvantages**: Implicit distribution (cannot compute $p(\mathbf{x})$); hard to train (no convergence guarantee, prone to mode collapse).

8.4 Diffusion Models

- **Forward Diffusion Process, 前向扩散**:
  - Given data $\mathbf{x}_0 \sim q(\mathbf{x})$, iteratively add Gaussian noise in T steps:

$$\mathbf{x}_t = \sqrt{1-\beta_t}\mathbf{x}_{t-1} + \sqrt{\beta_t}\epsilon_{t-1}$$

    where $\{\beta_t \in (0,1)\}_{t=1}^T$ is a variance schedule and $\epsilon_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
  - Transition probability:

$$q(\mathbf{x}_t|\mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\mathbf{x}_{t-1}, \beta_t\mathbf{I})$$

  - As $T \to \infty$, $\mathbf{x}_T$ becomes an isotropic Gaussian.
- **Closed-Form Sampling at Arbitrary Substep t**:
  - Let $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{s=1}^t \alpha_s$.
  - Recursive derivation:

$$\begin{aligned}\mathbf{x}_t &= \sqrt{\alpha_t}\mathbf{x}_{t-1} + \sqrt{1-\alpha_t}\epsilon_{t-1} \\ &= \sqrt{\alpha_t\alpha_{t-1}}\mathbf{x}_{t-2} + \sqrt{1-\alpha_t\alpha_{t-1}}\bar{\epsilon}_{t-2} \\ &= \cdots \\ &= \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon\end{aligned}$$

    where $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$.
  - Conditional distribution of $\mathbf{x}_t$ given $\mathbf{x}_0$:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I})$$

    (人话：x_t 就是 x_0 和随机噪声 ϵ 的一个加权平均)
- **Reverse Diffusion Process，反向去噪**:
  - The true reverse process $q(\mathbf{x}_{t-1}|\mathbf{x}_t)$ is unknown. If β_t is small enough, it is also Gaussian.
  - We train a neural network p_θ to approximate it:

$$p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t))$$

  - **Tractable Reverse Conditional Probability**: When conditioned on $\mathbf{x}_0$, the reverse probability has a closed form:

$$q(\mathbf{x}_{t-1}|\mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t\mathbf{I})$$

    where the parameters are:

$$\tilde{\beta}_t = \frac{1-\bar{\alpha}_{t-1}}{1-\bar{\alpha}_t}\beta_t, \quad \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1-\bar{\alpha}_{t-1})}{1-\bar{\alpha}_t}\mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}}\beta_t}{1-\bar{\alpha}_t}\mathbf{x}_0$$

- **Training Objective(optional)**:
  - Minimize the variational bound on negative log-likelihood:

$$\begin{aligned}\mathbb{E}[-\log p_\theta(\mathbf{x}_0)] &\leq \mathbb{E}_q\left[-\log \frac{p_\theta(\mathbf{x}_{0:T})}{q(\mathbf{x}_{1:T}|\mathbf{x}_0)}\right] \\ &= \mathbb{E}_q\left[-\log p(\mathbf{x}_T) - \sum_{t\geq 1}\log\frac{p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t)}{q(\mathbf{x}_t|\mathbf{x}_{t-1})}\right] \\ &= \mathbb{E}_q\Big[\underbrace{D_{KL}(q(\mathbf{x}_T|\mathbf{x}_0)\|p(\mathbf{x}_T))}_{L_T} \\ &\quad + \sum_{t>1}\underbrace{D_{KL}(q(\mathbf{x}_{t-1}|\mathbf{x}_t,\mathbf{x}_0)\|p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t))}_{L_{t-1}} - \underbrace{\log p_\theta(\mathbf{x}_0|\mathbf{x}_1)}_{L_0}\Big]\end{aligned}$$

  - For L_{t-1}, using reparameterization:

$$\begin{aligned}L_{t-1} &= \mathbb{E}_q\left[\frac{1}{2\sigma_t^2}\|\tilde{\mu}_t(\mathbf{x}_t,\mathbf{x}_0) - \mu_\theta(\mathbf{x}_t,t)\|^2\right] + C \\ &= \mathbb{E}_{\mathbf{x}_0,\epsilon}\left[\frac{1}{2\sigma_t^2}\left\|\frac{1}{\sqrt{\alpha_t}}\left(\mathbf{x}_t(\mathbf{x}_0,\epsilon) - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon\right) - \mu_\theta(\mathbf{x}_t(\mathbf{x}_0,\epsilon),t)\right\|^2\right] + C \\ &= \mathbb{E}_{\mathbf{x}_0,\epsilon}\left[\frac{\beta_t^2}{2\sigma_t^2\alpha_t(1-\bar{\alpha}_t)}\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2\right] + C\end{aligned}$$

  - **Simplified Objective** (Optimization via SGD):

$$L_{\text{simple}}(\theta) := \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2\right]$$

  (反正是 optional, 摆了, 当个吉祥物)

- **Advantages & Disadvantages**:
  - **Pros**: Higher generation quality than VAE/GAN; Explicit probability distribution.
  - **Cons**: Time-consuming training; Slow sampling generation due to large T (e.g., $T = 1000$).

9 Lec8: Foundation Models

9.1 Introduction

- **Definition**: Large-scale pre-trained models serving as a general-purpose basis for downstream tasks.
- **Adaptation**: Trained on vast data, fine-tuned for specific applications via transfer learning, prompting, or few-shot learning.
- **Applications**: Key to modern NLP (e.g., BERT, GPT-3, CLIP) and CV (e.g., DALL·E, Stable Diffusion, BEiT, SORA).
- **Architecture**: Built on multiple neural networks like MLP, CNN, RNN, and Transformer.

9.2 BERT (2018)

- **Concept**: Bidirectional Encoder Representations from Transformers, designed for context understanding.
- **Architecture**: Encoder-only Transformer.
- **Pre-training**: Masked Language Modeling (MLM) and Next Sentence Prediction (NSP).
- **Data/Training**: Trained on BookCorpus and Wikipedia; Base (110M) and Large (340M) versions.
- **Fine-tuning**: Adapt to specific tasks (e.g., Sentiment Analysis, QA) with fewer resources and smaller datasets.
- 一句话就是，BERT 只有 Transformer 的 Encoder 部分，专注于根据上下文理解文本。核心用处是可以判断某个词在当前语境的意思，或者句子之间的关系。

9.3 GPT-3 (2020)

- **Concept**: Large language model by OpenAI, capable of human-like text generation and diverse NLP tasks.
- **Architecture**: Decoder-only Transformer; 175B parameters.
- **Pre-training**: Self-supervised learning via Next Word Prediction (causal language modeling).
- **Data**: Trained on hundreds of billions of words (Common Crawl, WebText2, Books, Wikipedia).
- **Capabilities**: Strong zero/few-shot performance in coding (CSS, Python, etc.), translation, and summarization.
- 一句话就是，GPT-3 只有 Transformer 的 Decoder 部分，专注于根据前文生成文本。训练方式是预测下一个词是什么，可以用来续写文本，或者根据提示生成文本。

9.4 CLIP (2021)

- **Concept**: Contrastive Language-Image Pre-training by OpenAI.
- **Objective**: Align text and image representations in a shared vector space, where similar pairs are close and dissimilar ones are far apart.
- **Architecture**: Consists of a **Visual Encoder** (ViT or ResNet) and a **Text Encoder** (Transformer).
- **Training**: Contrastive pre-training on "WebImageText" (WIT) dataset with 400M image-caption pairs.
- **Loss Function**: Optimized via a symmetric cross-entropy loss over the similarity matrix:

$$L = -\frac{1}{N}\sum_i \ln \frac{e^{v_i^T w_i/T}}{\sum_j e^{v_i^T w_j/T}} - \frac{1}{N}\sum_j \ln \frac{e^{v_j^T w_j/T}}{\sum_i e^{v_i^T w_j/T}}$$

- **Applications**: Image captioning, text-to-image generation, and zero-shot prediction (using label text as prompts).
- 一句话：通过对比学习将文本和图像映射到同一特征空间。训练时让匹配的图文对向量尽可能接近，不匹配的远离。

9.5 Stable Diffusion (2022)

- **Concept**: A deep learning, text-to-image model by Stability AI based on Latent Diffusion Models (LDM).
- **Architecture**: VAE + UNet + CLIP + Diffusion.
- **Process**:
  - **Pixel Space**: Image x is encoded into latent space z by VAE Encoder $\mathcal{E}$, and decoded back to pixel space $\tilde{x}$ by Decoder $\mathcal{D}$.
  - **Latent Space**: The diffusion process (adding noise) and denoising process occur here.
  - **Denoising U-Net** ϵ_θ: Predicts and removes noise using cross-attention mechanisms (Q, K, V) and skip connections.
  - **Conditioning**: External guidance (Text, Semantic Map, Images) is injected via a conditioning encoder τ_θ into the U-Net.
- **Scale**: Version XL contains 3.5 billion parameters.
- 一句话：在潜空间 (Latent Space) 进行扩散和去噪的文生图模型，利用 VAE 降维以提高效率，并通过 Cross-Attention 引入文本引导。

9.6 LoRA: Low-Rank Adaptation

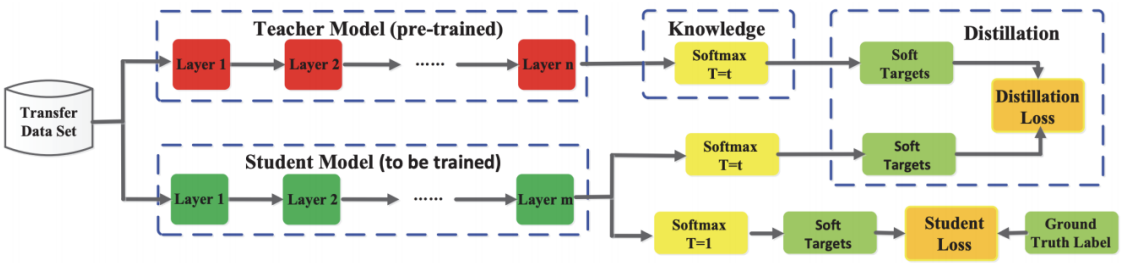
- **Concept**: Efficiently fine-tune large pre-trained models by decomposing the weight update matrix into two low-rank matrices.
- **Mathematical Formulation**:
  - For a pre-trained weight matrix $W \in \mathbb{R}^{d\times d}$, the output is modified as: $h = Wx + \Delta Wx = Wx + ABx$
  - where $A \in \mathbb{R}^{d\times r}$ and $B \in \mathbb{R}^{r\times d}$ are trainable parameters, with rank $r \ll d$.
  - Full-parameter update loss: $\mathcal{L} = \frac{1}{2}\|Y - WX\|_F^2$, gradient complexity $\mathcal{O}(d^3)$.
  - LoRA loss: $\mathcal{L} = \frac{1}{2}\|Y - (W + AB)X\|_F^2$.
- **Optimization**:
  - Gradients: $\frac{\partial \mathcal{L}}{\partial A} = -(Y - WX - ABX)(BX)^T$ and $\frac{\partial \mathcal{L}}{\partial B} = -A^T(Y - WX - ABX)X^T$.
  - Complexity: Total time complexity reduced to $\mathcal{O}(d^2r + r^2d)$, which is significantly less than $\mathcal{O}(d^3)$ when $r \ll d$.
- 一句话：不改变预训练权重 W，只训练降维后的 A 和 B。极大地减少了计算开销和显存占用。

9.7 Other Fine-tuning Strategies

- **Partial Fine-tuning**: Fine-tune only a subset of parameters (e.g., top layers or specific layers) while freezing the rest.
- **Multi-task Fine-tuning**: Train the model on multiple tasks simultaneously to improve generalization.
- **Adversarial Fine-tuning**: Incorporate adversarial examples to enhance model robustness.

9.8 Model Compression

- **Knowledge Distillation，知识蒸馏**: A pre-trained Teacher model guides the training of a smaller Student model, transferring generalization ability at a reduced scale.
  - **Loss function**: $\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \cdot T^2 \cdot \mathcal{L}_{\text{KL}}$
  - **Student loss (CE)**: $\mathcal{L}_{\text{CE}} = -\sum_i y_i \log(q_i)$, where y is the ground-truth label and q is the Student's predicted probability, $q_i = \frac{\exp(z_i/T)}{\sum_j \exp(z_j/T)}$.
  - **Distillation loss (KL)**: $\mathcal{L}_{\text{KL}} = \sum_i p_i \log\left(\frac{p_i}{q_i}\right)$, where p is the Teacher's predicted probability (soft target at temperature $T = t$).
  - **Temperature** T: A higher T smooths the softmax output, letting the Student learn inter-class similarity (dark knowledge). Use $T = 1$ at inference.
  - **Pros**: Significantly reduces model size while retaining the Teacher's generalization ability.
  - **Cons**: Relies on a high-quality Teacher and requires additional training time.



- **Quantization，量化**: Convert model weights and activations from floating-point to lower-precision formats to reduce memory and accelerate inference.
  - **Formats** (bits = sign + exponent + mantissa):
    - `fp32`: 1 + 8 + 23 bits
    - `fp16`: 1 + 5 + 10 bits
    - `bfloat16`: 1 + 8 + 7 bits (same exponent range as fp32, reduced precision)
    - `int8`: 8 bits integer; `int4`: 4 bits integer
  - **Example** (3.12345): `fp16` → `0 10000 1000111110`; `int4` → `0011`
  - **Pros**: Greatly reduces memory usage and accelerates inference.

  - **Cons**: Aggressive quantization may lead to significant accuracy loss.

- **Factorization，分解**: Decompose large weight matrices into smaller ones.
  - **Pros**: Mathematically sound; significantly reduces parameters.
  - **Cons**: Decomposition may introduce errors, requiring retraining to recover performance.

- **Pruning，剪枝**: Remove redundant parameters or structures to simplify the network.
  - **Pros**: Significantly reduces parameters and computation; may improve generalization.
  - **Cons**: Unstructured pruning requires specialized hardware to support sparse computation.

9.9 Challenges

- **Introduction**
  - **Hallucination**: LLMs occasionally generate plausible but incorrect content that (1) diverges from user's input, (2) misaligns with established knowledge, or (3) contradicts previously generated context.
  - **Interpretability**: Difficult to understand why a model produces a given output.
  - **Limited context length**: Cannot process arbitrarily long inputs.
  - **High compute requirements**: Training and inference demand significant hardware resources.

- **Hallucination Detection**
  - **Perplexity (information theory)**: Uncertainty measure for a discrete distribution: $\mathrm{PPL}(p) = b^{-\sum_x p(x)\log_b p(x)}$, where $b \in \{2, 10, e, \ldots\}$.
  - **Perplexity (LLM)**: For a token sequence $\mathbf{x} = (x_1, \ldots, x_n)$, an LLM with parameters θ models $p_\theta(x_i \mid \mathbf{x}_{<i})$. Perplexity is:

$$\mathrm{PPL}_\theta(\mathbf{x}) = \exp\left(-\frac{1}{n}\sum_{i=1}^{n}\log P_\theta(x_i \mid \mathbf{x}_{<i})\right)$$

    It quantifies the model's uncertainty when encountering new tokens.
  - **Applications of** PPL_θ:
    * **Training signal**: Minimizing PPL ≡ minimizing cross-entropy loss.
    * **Benchmarking**: Compare models on standard datasets (e.g., WikiText-2); lower PPL is better.
    * **Domain check**: Identify which base model best fits a target domain before fine-tuning.
    * **Fluency filter**: Unnatural sentences yield high PPL.
  - **Limitations of PPL**: Measures statistical fluency, not factual accuracy —a model can achieve low PPL while generating false statements.
  - **EigenScore** [Chen et al., 2024]: $E = \frac{1}{K}\sum_i^K \log(\lambda_i)$, where λ_i are eigenvalues; captures semantic consistency across multiple generated responses.

(可恶，还是看不懂吗)

10 Lec9: AI for Science

10.1 AI for Physics

- **Physical Laws and PDEs**: Most fundamental physical laws are expressed as PDEs. General form:

$$F\left(x, y, \ldots, t,\, u,\, \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y}, \ldots, \frac{\partial^2 u}{\partial x^2}, \frac{\partial^2 u}{\partial y^2}, \frac{\partial u}{\partial t}, \frac{\partial^2 u}{\partial t^2}, \ldots\right) = 0$$

  ODEs are special, simplified cases. Examples:

  - **Newton's Second Law**: $F = m\dfrac{d^2x}{dt^2}$
  - **Heat Transfer**: $\dfrac{\partial u}{\partial t} = \alpha\nabla^2 u$, where $u(x, y, z, t)$ is temperature, $\nabla^2 u = \dfrac{\partial^2 u}{\partial x^2} + \dfrac{\partial^2 u}{\partial y^2} + \dfrac{\partial^2 u}{\partial z^2}$ (Laplacian, spatial curvature), α is thermal diffusivity.
  - **Schrödinger Equation**: $i\hbar\dfrac{\partial \psi}{\partial t} = -\dfrac{\hbar^2}{2m}\nabla^2\psi + V\psi$, where $\psi(x, y, z, t)$ is the wave function (probability amplitude), $i\hbar\frac{\partial \psi}{\partial t}$ is time evolution, $-\frac{\hbar^2}{2m}\nabla^2\psi$ is kinetic energy, $V\psi$ is potential energy.
  - **Fluid Mechanics**: Navier-Stokes Equations.
  - **Electromagnetism**: Maxwell's Equations.

- **Solving PDEs**:

  - **Simple PDEs** (e.g., Heat Conduction): analytical solutions, e.g., separation of variables.
  - **Classical methods** (complex PDEs): FEM, FVM, FDM. Limitation: require manual mesh creation; computationally intractable for high-dimensional problems.
  - **Modern AI-based methods**: Neural Operators, Physics-Informed Neural Networks (PINNs). Advantages: mesh-free, overcome the curse of dimensionality, integrate physical laws with sparse data for inverse problems.

- **Physics-Informed Neural Networks (PINNs)**: Universal function approximators that embed physical laws (PDEs) into the learning process. General PDE form:

$$\mathcal{F}(u(z); \gamma) = f(z) \quad z \in \Omega, \qquad \mathcal{B}(u(z)) = g(z) \quad z \in \partial\Omega$$

  Ω: domain $\subset \mathbb{R}^d$ with boundary $\partial\Omega$

  z: $:= [x_1, \ldots, x_{d-1}; t]$, space-time coordinate

  u: unknown solution

  γ: physics parameters

  f: source data function

  $\mathcal{F}$: nonlinear differential operator

  $\mathcal{B}$: boundary/initial condition operator

  g: boundary function

- **Forward problem**: find u for every z (given γ).
- **Inverse problem**: determine γ from data.
- **PINN approximation** [Cuomo et al., JPC 2022]: a neural network $\hat{u}_\theta(z) \approx u(z)$ is trained by minimising:

$$\min_\theta \ \omega_{\mathcal{F}}\mathcal{L}_{\mathcal{F}}(\theta) + \omega_{\mathcal{B}}\mathcal{L}_{\mathcal{B}}(\theta) + \omega_d\mathcal{L}_{\text{data}}(\theta)$$

where $\mathcal{L}_{\mathcal{F}}$ enforces the PDE residual, $\mathcal{L}_{\mathcal{B}}$ enforces boundary/initial conditions, $\mathcal{L}_{\text{data}}$ fits observed data, and $\omega_{\mathcal{F}}, \omega_{\mathcal{B}}, \omega_d$ are loss weights.

  * $\mathcal{L}_{\mathcal{F}}(\theta) = \dfrac{1}{N_c}\displaystyle\sum_{i=1}^{N_c} \|\mathcal{F}(\hat{u}_\theta(z_i)) - f(z_i)\|^2$ (collocation points $\{z_i\}_{i=1}^{N_c}$)
  * $\mathcal{L}_{\mathcal{B}}(\theta) = \dfrac{1}{N_B}\displaystyle\sum_{i=1}^{N_B} \|\mathcal{B}(\hat{u}_\theta(z_i)) - g(z_i)\|^2$ (boundary points)
  * $\mathcal{L}_{\text{data}}(\theta) = \dfrac{1}{N_d}\displaystyle\sum_{i=1}^{N_d} \|\hat{u}_\theta(z_i) - u_i^*\|^2$ (observations $\{(z_i, u_i^*)\}_{i=1}^{N_d}$)

- **Example: PINN for Wave Equation**. 2D acoustic wave equation:

$$\rho\nabla\cdot\left(\frac{1}{\rho}\nabla u\right) - \frac{1}{v^2}\frac{\partial^2 u}{\partial t^2} = -\rho\frac{\partial^2 f}{\partial t^2}$$

where $u(t,x)$: pressure (wave-field); $f(t,x)$: point source; $v = \sqrt{\kappa/\rho}$: medium velocity; ρ: density; κ: adiabatic compression modulus. When ρ is constant and f negligible:

$$\nabla^2 u - \frac{1}{v^2}\frac{\partial^2 u}{\partial t^2} = 0$$

  - **Model** ($\hat{u}_\theta$): MLP, 10 layers of width 1024, softplus activations (hidden), linear output. Inputs: $x^{(1)}, x^{(2)}, t$; output: u.
  - **Loss**: $L(\theta) = L_p(\theta) + \lambda L_b(\theta)$, where

$$L_p(\theta) = \frac{1}{N_p}\sum_i \left(\left[\nabla^2 - \frac{1}{v(x_i)^2}\frac{\partial^2}{\partial t^2}\right]\hat{u}_\theta\left(x_i^{(1)}, x_i^{(2)}, t_i\right)\right)^2$$

$$L_b(\theta) = \frac{1}{N_b}\sum_j \left(\hat{u}_\theta\left(x_j^{(1)}, x_j^{(2)}, t_j\right) - u_{\text{FD}}\left(x_j^{(1)}, x_j^{(2)}, t_j\right)\right)^2$$

    L_p: N_p samples with $x_i \in [0, X_{\max}]$, $t_i \in [0, 0.2]$; L_b: N_b samples with $x_j \in [0, X_{\max}]$, $t_j \in [0, 0.02]$.

10.2 AI for Chemistry

- **Molecular Properties**: Determine a molecule's real-world function, safety, and value (e.g., drug efficacy, material strength).
  - **Typical properties**: Atomization Energy, Free Energy, Heat Capacity, Dipole Moment, Nuclear Magnetic Resonance, Binding Affinity, …
  - **Typical methods**:
    * Experimental Measurement: slow and expensive.
    * Quantum Chemistry (e.g., Density Functional Theory (DFT)): computationally intensive.

- **GNN for Molecular Property Prediction**: Input is a molecular graph G with node features x_v and edge features e_{vw}.
  - **Message passing** (hidden layers):

$$m_v^{t+1} = \sum_{w \in N(v)} M_t\big(h_v^t,\, h_w^t,\, e_{vw}\big)\,, \qquad h_v^{t+1} = U_t\big(h_v^t,\, m_v^{t+1}\big)$$

    where M_t (message) and U_t (update) are learnable functions.
    Example: $M_t(h_v^t, h_w^t) = c_{vw} h_w^t$, $c_{vw} = (\deg(v)\deg(w))^{-1/2} A_{vw}$; $U_t^v(h_v^t, m_v^{t+1}) = \mathrm{ReLU}(W^t m_v^{t+1})$.
  - **Output**: $\hat{y} = R\big(\{h_v^T \mid v \in G\}\big)$, where R is a readout function.
- **Molecule Graph Generation**: Computationally design novel molecular structures to explore chemical space. Typical deep learning methods:
  - Autoregressive models [Liu et al., 2018; Mercado et al., 2021]
  - Motifs-based models [Jin et al., 2020; Maziarz et al., 2022]
  - VAEs and GANs [Zhu et al., 2022]
  - Diffusion models [Vignac et al., 2023]

10.3 AI for Biology

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